

Product Backlog

Backlog ID	Epic/Story Title	User Story (if applicable)	Priority	Story Points	Acceptance Criteria Example	Sprint Target
EPIC-1	Familiarization with Existing Software	As a developer, I want to access and review the existing Python-based image-guided intervention software, so that I can understand its architecture and performance.	High	8	- Existing software is accessed and reviewed - Software is set up - Code runs successfully - Architecture documented	Sprint 1
STORY-1.1	Setup and Documentation	As a developer, I want to set up the existing software in a controlled environment, so that I can run and test it.	High	5	- Environment configured - System runs without errors - Setup steps documented	Sprint 1
EPIC-2	Modularization and Extensibility	As a developer, I want the architecture to be modular and extensible, so that future enhancements can be efficiently integrated.	High	13	Refactored components with clear interfaces	Sprint 1
STORY-2.1	Refactor Modules	As a developer, I want to refactor the code into independent modules, so the system is easier to extend.	High	8	- Independent modules - Code compiles - Functionality remains the same	Sprint 1
STORY-2.2	Tests	As a developer, I want to add unit and integration tests, so that the functionality is verified.	High	5	- Tests cover all functionality - Code passes all tests	Sprint 1
EPIC-3	2D Images to 3D Volume Reconstruction	As a surgeon, I want accurate 3D reconstruction from 2D ultrasound images, so that I am able to make better surgical decisions.	High	13	3D volume builds successfully from 2D images with improved efficiency and accuracy	Sprint 2
STORY-3.1	Analyze Current Code (Python and C++)	As a developer, I want to thoroughly analyze and review the existing reconstruction code, so that I can identify inefficiencies.	High	5	- Inefficiencies are documented - Areas are flagged for optimization	Sprint 2
STORY-3.2	Prototype Optimization	As a developer, I want to prototype an optimized reconstruction pipeline, so that performance is improved.	High	8	Prototype runs faster and produces equivalent or better 3D results	Sprint 2